20.2 Prokaryotes

Friday, October 16, 2026 1:17 PM

- include bacteria, archaea
- single celled organisms that can fully survive on their own
- abundant in air, water, and other places
- pro karyote → before nucleus
  - before, kernel
- size (1–10 μm) in length, 0.7 μm – 1.5 μm in width

- 20.2.1 Pasteur experiments
  - before him people thought things rotted due to spontaneous generation of mold, bacteria, maggots
  - He showed if you sterilize broth → no mold, then expose to indoor air → little mold, outdoor air → lot mold.
- 20.2.2 Prokaryote structure
  - Cell envelope
    - Plasma membrane
    - Cell wall

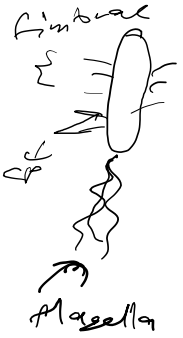
Prokaryotic cell
- Cytoplasm
  - Nucleoid
  - glycocalyx
  - Ribosome
  - Thylakoids
- Appendages
  - flagella
  - fimbrae
  - conjugation pilus

- Lack nucleus or organelles with membrane
- 1–10 μm by 0.7–1.5 μm wide
- Cell envelope
  - consists of plasma membrane cell wall and optionally glycocalyx/S-layer
  - plasma → encloses cyto, thin, semipermeable controls flow in/out in all animal/plant cells
  - cell wall → helps cell keep shape prevents it from exploding
  - glycocalyx → polysaccharide coating over wall
    - solid structure → called capsule
    - otherwise slime
  - S-layer → layer of proteins in place of glycocalyx
- Cytoplasm: includes the fluid (cytosol), and organelles inside except nucleus
  - nucleoid region → 1 chromosome of

circular dna some bacteria may have 1–4 copies

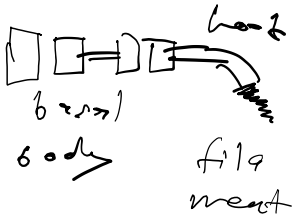
— many have accessory rings of DNA called plasmids, which code for antibiotic resistance toxins etc.

— Appendages
  — fimbrae: hair like bristles, allow adhesion to surfaces (gonnarhea bacteria uses this)
  — conjugation pilus → a thin, hollow appendage that allow transferring DNA to other cells
    (pilus = hair)



— flagella
  — rotating filament (strands) that propels cell
  — consists of
    filament made of protein flagellin wound in helix
    inserted into hook (that is what it's called)
    anchored at basal (at base) body (rotating structure)
    (part which rotates, causing flagella to rotate.)

- archaeflagella like this, but no basal body, thinner hook



Extra

- nucleoid
  - location of bacterial chromosome in cytoplasm
- plasmid
  - extrachromosomal ring of DNA

- Reproduction in prokaryotes
  - do via binary fission
    - parent and daughter cells have same DNA
    - 1. the circular DNA replicates
    - 2. New Plasma cell wall created
    - 3. separates
  - generation time - 12 minutes
  - thus mutate rapidly

- Prokaryotes are haploid
  - haploid: 1 set of chromosomes
  - Eukaryotes like humans
    - → most cells diploid 2 sets

of chromosomes | from each parent

- exception are sperm and egg cells (duh!)

- Genetic Recombination in Prokaryotes
  - Conjugation
    - 2 bacteria linked by conjugation pilus
    - DNA passed from Donor to Recipient
  - Transformation
    - A prokaryote picks up free pieces of DNA, incorporates into genome
  - Transduction
    - bacteriophages carry DNA b/w cells
    - sorta like mozzies and disease.

Extra:

Prokaryote
- bacteria
- archae

Eukaryote
- fungi
- protists
- plants
- animals

20.3 Bacteria

- Most common type of prokaryote
- Cell wall is made out of Peptidoglycan
  - polysaccharides linked by amino acids
- Type of Bacteria, based on cell wall
  - Gram Positive
    - thick peptidoglycan wall
    - no plasma layer outside wall
  - Gram Negative
    - thin peptidoglycan wall
    - 2nd plasma surrounding

— Cell plasma surrounding wall makes it antibiotic resistant.

— Gram staining process

1. Apply crystal violet
   enters into cell for both
   makes both look purple
2. Apply Gram's Iodine
   forms violet–iodine complex
   in both, complex to big to leave
3. Apply Alcohol wash

— For gram positive, thick peptidoglycan wall keeps complex, stays purple (wall dehydrated less permeable)

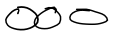
— Negative: alcohol disrupts outer plasma washes away complex, looks unstained

4. Safranin (red dye)

— Positive: Purple masks red, still purple

— Negative: looks red now

— Bacteria Shape/ Classification

- Spirilli (spiral/helical/spring)
- Cocci (spherical) 
- Bacilli (rod-shaped)
- Patterns of growth
  - strept → chains
  - diplo → pairs
  - staph → clusters

- Bacterial Metabolism

obligate aerobes
- need oxygen to grow

obligate anaerobes
- can only grow if there is no oxygen

(botulism, gangrene, tetanus)

facultative anaerobes → can grow with or without oxygen

- autotroph → create own food
- photoautotrophs - create own food like plant from light

↓ light ↓ self ↓ food

- use solar energy to reduce carbon dioxide to organic compounds
- carry out photosynthesis (both kinds)
- types: anoxygenic and oxygenic

- anoxygenic
  - Ex: green sulfur bacteria
    purple bacteria
  - live in O_2 poor
  - can't photosynthesize if O_2 present
  - don't release O_2
  - only use photosystem I
    ↓
    doesn't require water
- oxygenic
  - acts like plants
  - contain chlorophyll a
  - releases O_2
  - Ex: cyanobacteria
- Chemoautotrophs
  - make food by self using chemicals

— photoautotrophs use light energy, these oxidize inorganic compounds to get energy, both convert CO_2 to organics

— Ex: H_2, NH_3, H_2S

— Instead of photosynthesis, carry out chemosynthesis

— Ex: deep-sea bacteria use sulfur compounds, nitrifying bacteria can vert NH_3 to NO_3^- and NO_2^-

— Heterotrophic Bacteria
- different
- food

— get food from organic nutreits produced by others

— also called chemoheterotrophs

— saprotrophs — decomposers

— Symbiotic relationships

— means two species live together in an intimate way

types:

- metalism:
  - both benifit
  - Ex: nitrogen fixing bacteria, and (soy & corn, alfalfa, clover) (live in root nodules)
  - Ex: Bacteria in our stomach make B_{12} & K vitamins
  - cellulose digesting intestinal bacteria
- Commensalism
  - one bacteria makes life better for another (gets nothing in return)
  - anaerobic bacteria in stomach survive as E. coli uses up O_2
- parasitic bacteria: pathogens
  - gram negative bacteria release superantigen lipopolysaccharide on antibiotic death, causes excess immune response

- on rusty nail, cloistrum tetani produces toxin which causes tetanics (causes excess muscle contraction)

- Endospores
  - Bacteria can hibernate form spores made from cytoplasm and copy of chromosomes
  form endospore coat, survive extreme conditions, once conditions good again they germinate back into normal bacteria. Ex: botulism food poisoning can survive in tin cans germinates with water

- Antibiotics